



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 • STANDARD 6.NOS.4

# Least Common Multiple

Lesson 6-12 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 3 Enrichment



### TODAY'S OBJECTIVES

**Content Objective:** I can find the least common multiple (LCM) of two numbers by listing or comparing their multiples.

**Language Objective:** I can explain how I found the LCM using the words multiple, common multiple, skip counting, and least common multiple.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Least Common Multiple

Multiple

Common multiple

Skip counting

Prime factorization



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The smallest number that is a multiple of both is their

.

2

The product of a number and any whole number is a

of that number.

3

A multiple that two or more numbers share is a  
multiple.

4

Counting forward by a number other than 1, like 6, 12, 18, is called

.

5

Writing a number as  numbers multiplied together is its prime factorization.

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**How does the LCM help you figure out the fewest hot dogs and buns to buy so they come out even?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Least Common Multiple** — The least positive number that is a multiple of two or more numbers.

*Mínimo común múltiplo — El menor número positivo que es múltiplo de dos o más números.*

- ☐ **Multiple** — What you get when you multiply a number by 1, 2, 3, and so on.

*Múltiplo — Lo que obtienes al multiplicar un número por 1, 2, 3, y así.*

- ☐ **Common multiple** — A number that two or more numbers both go into.

*Múltiplo común — Un número al que llegan dos o más números al multiplicar.*

- ☐ **Skip counting** — Counting by a number, like 2, 4, 6, to list its multiples.

*Conteo salteado — Contar de un número en un número, como 2, 4, 6, para listar sus múltiplos.*

- ☐ **Prime factorization** — Writing a number as prime numbers multiplied together. It helps you find the LCM.

*Factorización prima — Escribir un número como números primos multiplicados. Ayuda a encontrar el mcm.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**I need to buy \_\_\_\_ hot dogs and \_\_\_\_ buns because the LCM of 8 and 6 is \_\_\_\_.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** The next shared check is a common multiple of 4 and 6, not their sum.

**Evidence:** Students reason that each astronaut repeats on multiples of their own number, so they only overlap at a common multiple, and adding  $4 + 6$  does not land on both schedules.

**Reasoning:** This evidence connects the definition of least common multiple to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- I need to buy \_\_\_\_ hot dogs and \_\_\_\_ buns because the LCM of 8 and 6 is \_\_\_\_.
- My answer is \_\_\_\_.

*Mi respuesta es \_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.
- I know because \_\_\_\_.

*Mi afirmación es \_\_\_\_.*

*Lo sé porque \_\_\_\_.*

- So \_\_\_\_.

*Entonces \_\_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.



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## Answer Key & Teacher Guide

1. **Notes 1:** Least Common Multiple
2. **Notes 2:** Multiple
3. **Notes 3:** Common multiple
4. **Notes 4:** Skip counting
5. **Notes 5:** Prime factorization
6. **Watch for this mistake:** *A common mistake in Least Common Multiple is multiplying the two numbers together instead of listing multiples to find the smallest shared one. For example, for 4 and 6, a student calculates  $4 \times 6 = 24$  and calls that the LCM, but the actual least common multiple is 12 (multiples of 4: 4, 8, 12; multiples of 6: 6, 12 — 12 is the first number that appears in both lists). This shortcut only works when the two numbers share no common factors (like 4 and 9, where the LCM really is  $36 = 4 \times 9$ ); for numbers like 4 and 6 that share a factor of 2, multiplying overshoots the true LCM. Always list multiples of both numbers and pick the first one that appears in both lists.*
7. **Try It 1:** A. 12 — Skip count: multiples of 4 are 4, 8, 12; multiples of 6 are 6, 12. The first shared multiple is  $LCM(4, 6) = 12$  minutes.
8. **Try It 2:** A. 24 —  $LCM(8, 12) = 24$ . Multiples of 8: 8, 16, 24; multiples of 12: 12, 24. The first shared multiple is 24 minutes.